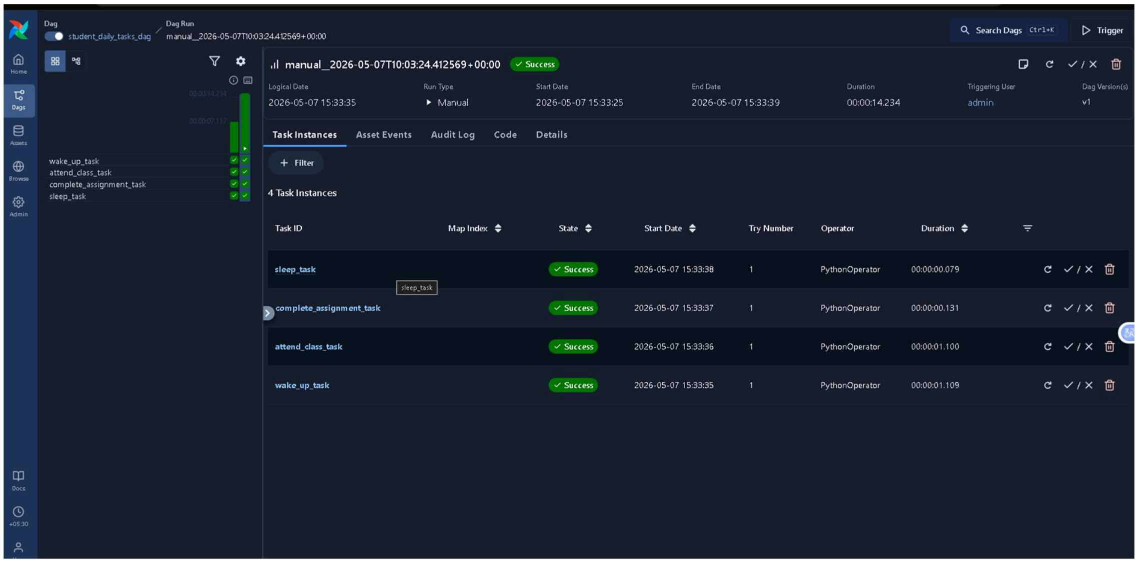



```
root@ubuntu:~$ docker exec -it airflow bash
airflow@ab161ec4145e:/opt/airflow$ cd /opt/airflow/dags
airflow@ab161ec4145e:/opt/airflow/dags$ ls
__pycache__  first_dag.py  student_dag.py
airflow@ab161ec4145e:/opt/airflow/dags$ cat /tmp/student_marks.txt
Math,80
Science,75
English,90
Python,95
airflow@ab161ec4145e:/opt/airflow/dags$ cat /tmp/result.txt
Student Result Summary
Total Marks = 340
Result = PASS
airflow@ab161ec4145e:/opt/airflow/dags$
```

Create and Execute Employee Salary DAG

1. Run Apache Airflow Container

```
docker run -d -p 8080:8080 --name airflow apache/airflow
standalone
```

---

2. Verify Running Container

```
docker ps
```

---

3. Retrieve Login Credentials

```
docker logs airflow
```

---

4. Access the Container

```
docker exec -it airflow bash
```

---

5. Navigate to DAGs Folder

```
cd /opt/airflow/dags
```

---

6. Create DAG File

```
cat > employee_salary_dag.py
```

Paste the code

```
from airflow import DAG
from airflow.operators.python import PythonOperator
from datetime import datetime
def create_employee_file():
    with open("/tmp/employees.txt", "w") as f:
        f.write("Rahul,45000\n")
        f.write("Sneha,52000\n")
        f.write("Amit,61000\n")
        f.write("Priya,47000\n")
        f.write("Kiran,39000\n")
def read_employee_data():
    with open("/tmp/employees.txt", "r") as f:
        for line in f:
            print(line.strip())
```

```python
def calculate_salary_expense():
    total = 0
    count = 0
    with open("/tmp/employees.txt", "r") as f:
        for line in f:
            salary = int(line.strip().split(",")[1])
            total += salary
            count += 1
    print(f"Total Salary Expense = {total}")
    return {"total": total, "count": count}

def find_highest_salary(ti):
    highest_salary = 0
    employee_name = ""
    with open("/tmp/employees.txt", "r") as f:
        for line in f:
            name, salary = line.strip().split(",")
            salary = int(salary)
            if salary > highest_salary:
                highest_salary = salary
                employee_name = name
    print(f"Highest Salary = {highest_salary}")
    print(f"Employee = {employee_name}")
    ti.xcom_push(key="highest_salary", value=highest_salary)
    ti.xcom_push(key="employee_name", value=employee_name)


def generate_salary_report(ti):
    data = ti.xcom_pull(task_ids="calculate_salary_expense")
    total = data["total"]
    count = data["count"]
    with open("/tmp/salary_report.txt", "w") as f:
        f.write("Employee Salary Report\n")
        f.write(f"Total Employees = {count}\n")
        f.write(f"Total Salary Expense = {total}\n")
        f.write("Status = Processed Successfully\n")

with DAG(
    dag_id="employee_salary_dag",
    start_date=datetime(2025, 1, 1),
    schedule="@daily",
    catchup=False
) as dag:

    t1 = PythonOperator(task_id="create_employee_file",
python_callable=create_employee_file)
```

```
    t2 = PythonOperator(task_id="read_employee_data",
python_callable=read_employee_data)
    t3 = PythonOperator(task_id="calculate_salary_expense",
python_callable=calculate_salary_expense)
    t4 = PythonOperator(task_id="find_highest_salary",
python_callable=find_highest_salary)
    t5 = PythonOperator(task_id="generate_salary_report",
python_callable=generate_salary_report)

    t1 >> t2 >> t3 >> t4 >> t5
```

---

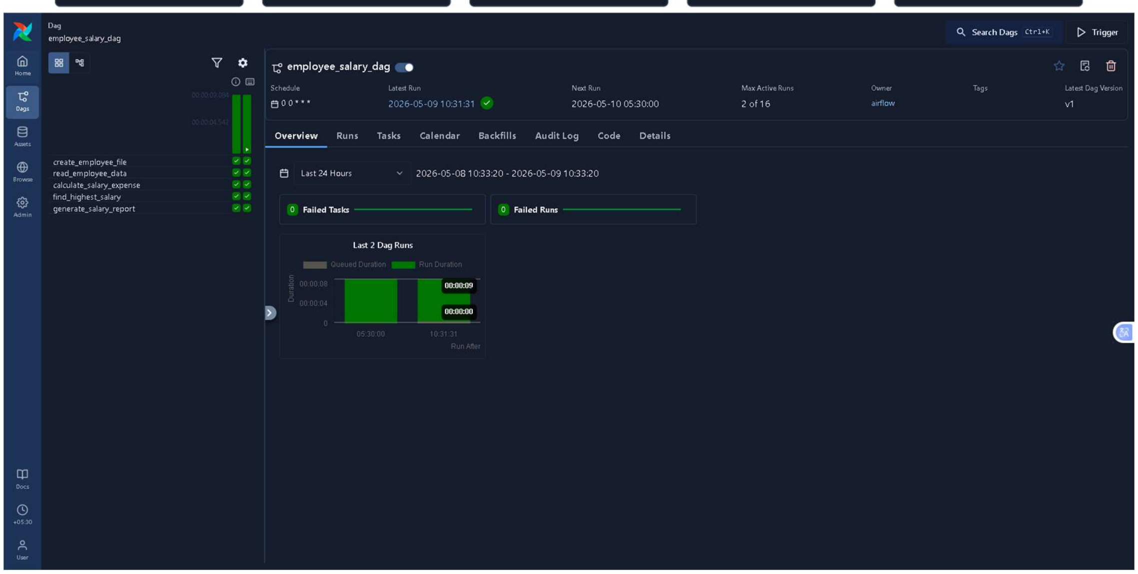
7. Verify File Creation

```
ls
```

---

8. Exit Container

```
exit
```




```
root@ubuntu:~$ docker exec -it airflow bash
airflow@00280c4c8bbf:/opt/airflow$ cd /opt/airflow/dags
airflow@00280c4c8bbf:/opt/airflow/dags$ cat /tmp/salary_report.txt
Employee Salary Report
Total Employees = 5
Total Salary Expense = 244000
Status = Processed Successfully
airflow@00280c4c8bbf:/opt/airflow/dags$
```